

Area Laws in Quantum Nash Equilibria

C. L

March 3, 2026

Abstract

Abstract here.

1 Setup

1.1 Quantum game formulation

We consider an L -player quantum game where each player i controls a single qubit. The joint state of all players is described by a quantum state $|\psi\rangle \in (\mathbb{C}^2)^{\otimes L}$. Each player's strategy consists of a single-qubit unitary $U_i \in \text{SU}(2)$ applied to their qubit. Given a strategy profile $(U_1, \dots, U_L)$ applied to a fiducial state $|\psi_0\rangle$, the resulting state is

$$|\psi\rangle = (U_1 \otimes U_2 \otimes \dots \otimes U_L) |\psi_0\rangle. \quad (1)$$

The payoff for player i is defined as the expectation value of a Hermitian operator H_i :

$$E_i = \langle \psi | H_i | \psi \rangle. \quad (2)$$

In the quantum Prisoner's Dilemma, the Hamiltonians H_i are diagonal in the computational basis,

$$H_i = \sum_{\mathbf{s} \in \{0,1\}^L} h_i(\mathbf{s}) |\mathbf{s}\rangle \langle \mathbf{s}|, \quad (3)$$

where $h_i(\mathbf{s})$ assigns a real-valued payoff to each measurement outcome $\mathbf{s} = (s_1, \dots, s_L)$.

1.2 Payoff structure

For $L = 3$ players, the payoff values are given in Table 1. The game is symmetric under cyclic permutation of the players:

$$h_i(\mathbf{s}) = h_{i+1 \bmod 3}(\sigma(\mathbf{s})), \quad (4)$$

where σ denotes the corresponding cyclic permutation of the bit string. Mutual cooperation ($\mathbf{s} = 000$) yields a payoff of 6 per player, while mutual defection ($\mathbf{s} = 111$) yields only 2—the characteristic tension of the Prisoner's Dilemma. A single defector receives 10 while the remaining cooperators receive 3 or 0, creating an incentive to deviate from cooperation.

Outcome $\mathbf{s}$	$h_1(\mathbf{s})$	$h_2(\mathbf{s})$	$h_3(\mathbf{s})$
000	6	6	6
001	3	3	10
010	3	10	3
011	0	6	6
100	10	3	3
101	6	0	6
110	6	6	0
111	2	2	2

Table 1: Payoff values for the 3-player quantum Prisoner's Dilemma.

For general L , players are arranged on a ring with pairwise interactions. Each player interacts with their two nearest neighbors via a 2-player Prisoner's Dilemma Hamiltonian $H^{(2)}$ with payoffs $(R, S, T, P) = (3, 0, 5, 1)$, and the total payoff for player i is the average over incident edges:

$$H_i = \frac{1}{2} \left(H_{i,i+1}^{(2)} + H_{i,i-1}^{(2)} \right), \quad (5)$$

where indices are taken modulo L .

1.3 Matrix product state representation

For large L , storing the full state vector $|\psi\rangle \in (\mathbb{C}^2)^{\otimes L}$ is intractable. We represent quantum states as matrix product states (MPS) with open boundary conditions:

$$|\psi\rangle = \sum_{s_1, \dots, s_L} A_1^{s_1} A_2^{s_2} \cdots A_L^{s_L} |s_1 s_2 \cdots s_L\rangle, \quad (6)$$

where each $A_j^{s_j}$ is a matrix of dimension $\chi_{j-1} \times \chi_j$ with $\chi_0 = \chi_L = 1$. The bond dimension $\chi = \max_j \chi_j$ controls the entanglement capacity of the ansatz. For $\chi = 1$, the state is a product state (zero entanglement); for $\chi = 2^{\lfloor L/2 \rfloor}$, the MPS can represent any state exactly.

In practice, we work with bond dimension $\chi = 2$, which is sufficient to capture states with moderate entanglement, including GHZ-type states $|\text{GHZ}\rangle = (|0\rangle^{\otimes L} + |1\rangle^{\otimes L})/\sqrt{2}$.

1.4 Nash equilibrium and exploitability

A state $|\psi\rangle$ constitutes an ε -Nash equilibrium if no player can increase their payoff by more than ε through a unilateral unitary deviation:

$$\max_{U_i \in \text{SU}(2)} \langle \psi_i(U_i) | H_i | \psi_i(U_i) \rangle - \langle \psi | H_i | \psi \rangle < \varepsilon \quad \forall i, \quad (7)$$

where $|\psi_i(U_i)\rangle$ denotes the state with U_i applied only to qubit i . The left-hand side defines the *exploitability* $\varepsilon_i(\psi)$ of player i .

We find Nash equilibria using *simultaneous differential best response dynamics*. At each iteration n , we compute the gradient of each player's payoff with respect to their local unitary:

$$M_i = \frac{\partial E_i}{\partial U_i} = \text{tr}_{j \neq i} [H_i |\psi\rangle\langle\psi|], \quad (8)$$

where the partial trace is over all qubits except i . The unitary update is extracted via the SVD of the matrix $\mathbb{I} - \alpha M_i$: if $\mathbb{I} - \alpha M_i = Y \Sigma Z^\dagger$, the update is $U_i = Z Y^\dagger$, which is the unitary closest to the gradient step in operator norm. All players update simultaneously, and the procedure is iterated until the total exploitability $\sum_i \varepsilon_i < \varepsilon_{\text{thr}}$.

The exploitability ε_i is computed by globally optimizing over $\text{SU}(2)$ using differential evolution, ensuring that the reported Nash equilibria are validated against arbitrary (not just infinitesimal) deviations.

2 Area laws

A quantum game is defined by the triple $(\hat{h}_i, G_i, N)$ where N is the number of players. We would like to study the quantum advantage associated with the underlying resource state.

- In the product state orbit, the game is purely classical (with automatic extension to mixed strategies by controlling the probability amplitudes of individual qubits)
- In the volume law cases, we get $\text{tr}(U_i \rho U_i^\dagger \hat{O}) \approx \text{tr} \rho \hat{O}$ for any local observable $\hat{O}$.
- So there might be an optimal entanglement structure for the underlying resource state for the maximal quantum advantage (in the sense of mechanism design).

2.1 Introducing non-commutativity

- Product state case: always find a Nash equilibrium irrespective of the payoff Hamiltonians.
- Adding entanglement by doing a multi-qubit unitary: can find NE for random Ising game with $\hat{h}_L = \hat{h}_0 + V(g_L)$, $\hat{h}_R = \hat{h}_0 + V(g_R)$, where $V(g) = g \hat{X}_i$ and we randomly pick g_L, g_R from a distribution.
- This statement breaks down when we consider the random Heisenberg interactions with $\hat{h}_i = n_x \hat{X}_i \hat{X}_{i+1} + n_y \hat{Y}_i \hat{Y}_{i+1} + n_z \hat{Z}_i \hat{Z}_{i+1}$. ($n_x^2 + n_y^2 + n_z^2 = 1$ chosen at random). Adding entanglement in the initial states let the interaction dynamics to end up in a limit cycle.
- Note that all games considered so far are translationally invariant, i.e. $T^\dagger \hat{h}_i T = \hat{h}_i$